Faruk Efe Yencilek

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Education

McMaster University **Hamilton, ON**
Bachelor of Computer Science, Honours Co-op, 3.73 GPA *September 2023 – May 2027*

- **Relevant Coursework:** Software Design (A+), Computer Networks & Security (A+), Cryptography (A+), Machine Learning (A-), Operating Systems (A), Data Structures & Algorithms (A-), Predictive and Intelligent Control, Real-Time Systems and Control Applications
- **Awards:** 3× Dean's Honour List, Engineering Award of Excellence ($3000)

Technical Skills

Languages: Python, TypeScript, C & C++, SQL, Rust
Full-stack: React, Next.js, Supabase, PostgreSQL, FastAPI, Axios, NextAuth, PyQt6
AI & ML: PyTorch, NumPy, pandas, Seaborn, scikit-learn, XGBoost, TensorFlow
DevOps & Systems: Git, Docker, Linux, Netlify

Experience

Artificial Intelligence Researcher **Hamilton, ON**
McMaster Society of Artificial Intelligence *September 2026 – Present*

- Selected for a research team of four to six students that takes an AI project from ideation to a final research paper and presentation at **CUCAI 2027**.
- Contributing research, literature review, and technical build work in weekly working sessions through March 2027.

AI Engineer Intern **Ankara, Türkiye**
HAVELSAN Inc. *August 2026 – September 2026*

- Built a custom probabilistic transformer model that predicts the future displacement of ships, a learnable alternative to Interacting Multiple Model (IMM) filters.
- Implemented an encoder-decoder backbone followed by a mixture density network that learns the Gaussian distribution of K different motion banks, optimized with a negative log loss (NLL).
- The resulting model performed equal to the IMM benchmark and the classical transformer benchmark on 10% of the dataset, with tanh gradient saturation estimated under 1%.
- Co-built a data-generation pipeline using Monte-Carlo and Latin Hypercube sampling methods, with a FastAPI backend service and Docker deployment hosting the pipeline for different model repositories.

Software Engineering Intern **Ankara, Türkiye**
ROKETSAN Inc. *June 2026 – July 2026*

- Built telemetry data monitoring software for the testing servers to diagnose failing test cases, using sockets and PyQt6.
- Wrote a module to choose the local interface to listen for any connection over TCP/UDP for the flight software telemetry data.
- Implemented a custom packet parser algorithm from bitstream data based on the user offset configuration, able to visualize the telemetry data of any sort.
- Wrote a CRC checker for packet verification, and a modernized KPI dashboard for the monitoring visuals.

Vice President **Hamilton, ON**
McMaster SynBio *December 2025 – Present*

- Led a 50-person student research team in the **iGEM 2025** and **iDEC 2026** biotechnology competitions.
- Assisted the entrepreneurship subtrack building its financial model and market analysis, helping the team raise **$50,000** in the Desjardins Cooperathon for the 2026 project.
- Contributed to the dry-lab modelling track, co-building a transformer model that uses a frozen RNA encoder backbone and a multi-layer head for structural inference.
- Developed and managed the team's project webpages, writing the deployment scripts, standardizing shared components for uniform development, and coordinating commits from five contributors on separate branches. Live for iGEM 2025 and iDEC 2026.

Projects

Lab Inventory Management System **May 2025 – July 2025**

- Full-stack inventory, ordering, and task-delegation tool for a 50-person lab, taken to a working proof of concept with a server-rendered **Next.js** and **TypeScript** front end.
- Designed the schema and database endpoints on Supabase-hosted **PostgreSQL**, with user session authentication, separate account types for sign-in, and **row-level security** keeping each role inside its own records.